

YVES ALAIN IRAGENA

Applied ML Engineer & Graduate Researcher · Washington D.C. · MS Data Analytics, May 2026 · F-1 CPT
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ML Engineer focused on **multimodal AI, LLM systems, and AI infrastructure**. Graduate Research Assistant at MAIL Lab (Catholic University of America) — active ICME 2026 Grand Challenge contributor. Builds end-to-end: research, backend engineering, and full-stack AI deployment. GPA: **3.7/4.0**.

EDUCATION

M.S. Data Analytics — Multimodal AI Focus · Catholic University of America, Washington D.C. Expected May 2026 GPA: 3.7/4.0 · RA

B.S. Computer Science — Honors & Distinction · KL University, Vijayawada, India 2019 – 2023 CGPA: 8.9/10

TECHNICAL SKILLS

AI / ML	PyTorch · Wav2Vec2 · BiLSTM · Attention · CNN · Scikit-learn · NumPy · pandas
LLM Systems	LangGraph · LangChain · MCP · RAG · Pinecone · FAISS · Chroma · Claude API · OpenAI API
Backend / Infra	Python · FastAPI · Node.js · PostgreSQL · Docker · Kubernetes · AWS · GitHub Actions
Frontend	React · Next.js · TypeScript · WebSockets · Tailwind CSS

EXPERIENCE

Graduate Research Assistant · MAIL Lab, Catholic University of America · Washington D.C. Jan 2025 – Present

- Developing **DPADD-ES**: mask-based architecture separating speech and environment representations for audio deepfake detection — **0.8923 macro-F1** on CompSpoonV2 (250K clips, 5 classes); submitted to **ICME 2026 Grand Challenge (ESDD2)**.
- Built adversarial RAG security framework with prompt shielding, semantic re-ranking, and output validation across Pinecone, FAISS, and Chroma — **82% hallucination reduction**.
- Implementing real-time synthetic speech classifier (Wav2Vec2 + BiLSTM + attention) for production audio integrity pipelines.

Data & Operations Analyst · Health Target Parapharmacy Ltd · Kigali, Rwanda Oct 2023 – Jan 2025

- Built Python and SQL automation pipelines eliminating 20+ hrs/week of manual data processing; **70% reduction** in data entry errors across multi-location retail operations.
- Designed and deployed inventory tracking system across the company's wholesale and import divisions.

SELECTED PROJECTS

DPADD-ES — Audio Deepfake Detection

[GitHub](#) 

[PyTorch](#) · [Wav2Vec2](#) · [BiLSTM](#) · [Attention](#) · [CompSpoonV2](#)

Mask separation model isolating speech artifacts from background — 0.8923 macro-F1 vs 0.63 baseline on unseen generators. ICME 2026 Grand Challenge (ESDD2).

AuraCommerce — Agentic BI Dashboard

[GitHub](#) 

[Python](#) · [FastAPI](#) · [OpenAI GPT-4o-mini](#) · [SQLite](#) · [Plotly.js](#) · [Prometheus](#)

3-agent pipeline (Analyst → Sentinel → Reporter) monitoring e-commerce KPIs in real time. Dual-mode Sentinel: GPT-4o-mini + rule-based fallback for zero-downtime reliability.

Gmail AI Semantic Autopilot

[GitHub](#) 

[Python](#) · [Claude Haiku](#) · [Anthropic SDK](#) · [FastAPI](#) · [Gmail API](#)

Daemon classifying emails into 7 intent categories — 98%+ accuracy, auto-archives cold outreach and newsletters, saves ~10 hrs/week at <\$0.01/day API cost.

Real-Time Ops Dashboard

[GitHub](#) 

[Next.js 16](#) · [TypeScript](#) · [Supabase Realtime](#) · [Tailwind CSS 4](#)

Live task board on Supabase postgres_changes subscriptions — DB writes on screen in <50ms. Optimistic UI with automatic rollback on write failure.

JP AutoHub SEO Engine

[GitHub](#) 

[Python](#) · [LangChain](#) · [AWS](#) · [PostgreSQL](#)

Multi-agent LangChain pipeline for SEO automation — keyword extraction, AI content generation, and publishing on a live auto-parts e-commerce platform.

Authorized to work in the U.S. under F-1 CPT — no sponsorship required for internship roles. Available Summer / Fall 2026.